Using Classification Trees for Predicting National Bridge Inventory Condition Ratings

Basak Aldemir Bektas[1]; Alicia Carriquiry[2]; and Omar Smadi[3]

Abstract: In bridge management practice, bridge condition is the fundamental information needed to allow decision makers to make well-informed decisions regarding preservation, rehabilitation, or replacement of a bridge or network of bridges. The National Bridge Inventory (NBI) condition ratings, collected since the early 1970s, and the Commonly Recognized (CoRe) element condition data, collected since the early 1990s, are two major sources of bridge condition data in the United States. General NBI condition ratings are utilized for performance assessment, performance reporting, resource allocation, and selection of bridge projects by all levels of government. Since the early 1990s, the bridge management community has been interested in an algorithm to predict the categorical NBI condition rating classes from the more quantitative and detailed CoRe element condition data. An algorithm with sufficient predictive accuracy would make only CoRe element inspections necessary and would provide time and resource savings. This paper presents a new methodology for this purpose, using classification and regression trees (CARTs). The CART analyses were conducted with the bridge condition data provided by three state transportation agencies, using data from 2006 to 2010. The statistical results point to a more accurate prediction method than the previous algorithms described in the literature. **DOI:** 10.1061/(ASCE)IS.1943-555X.0000143. *© 2013 American Society of Civil Engineers.*

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Introduction

Transportation agencies collect bridge condition data to assess the current condition and identify the future needs for maintenance, repair, rehabilitation, and reconstruction activities. These data are also used by the federal government to assess the health of the national bridge network and to allocate federal bridge funds among the states. Bridge condition data are also major inputs for project- and network-level programming models of bridge management systems (BMSs).

The National Bridge Inspection Standards (NBISs), implemented by the Federal Highway Administration (FHWA) in 1968 (Lichtenstein 1993), form the basis of bridge inspection and bridge management in the United States today. Typically, all bridges located on public roads with lengths more than 6.1 m (20 ft) are inspected at least once every 2 years. Data on 116 National Bridge Inventory (NBI) items are reported annually to FHWA by the state transportation agencies. These data are compiled into the NBI by FHWA, and condition and performance reports based on the NBI data are prepared and submitted to Congress (FHWA 2009).

[1]Research Scientist, Institute for Transportation, Iowa State Univ., Ames, IA 50011 (corresponding author). E-mail: basak@iastate.edu

[2]Professor, Dept. of Statistics, Iowa State Univ., Ames, IA 50011. E-mail: alicia@iastate.edu

[3]Research Scientist, Institute for Transportation and Adjunct Assistant Professor, Dept. of Civil, Construction, and Environmental Engineering, Iowa State Univ., Ames, IA 50011. E-mail: smadi@iastate.edu

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NBI items 58 through 66 in FHWA's Recording and Coding Guide (FHWA 1995) constitute the NBI condition ratings. Three of them are especially important and more frequently utilized by the bridge management community because they directly affect the Highway Bridge Program (HBP) funding eligibility criteria. Assigned to the three major parts in a bridge structure, these NBI condition ratings are deck, superstructure, and substructure condition ratings (items 58, 59, and 60, respectively). Deck is the roadway portion of a bridge that directly carries the traffic load (Fig. 1). The superstructure is composed of all structural components, such as beams, girders, and bearings, which receive the loads from the deck and transfer them to the substructure. All structural elements below bearings, such as columns, piers, piles, and footings, which support the span of a bridge, constitute the substructure. The NBI condition ratings are assigned on a scale of 0 to 9 and according to the specifications in the Recording and Coding Guide.

In addition to the NBI condition data, many state departments of transportation (DOTs) also collect element-level condition data to use in their BMSs. Pontis BMS, the product of a research project initiated by FHWA and licensed by AASHTO, is the predominant BMS used by 44 state agencies in the United States (Bektas 2011). The norm in Commonly Recognized (CoRe) element definitions and inspections in the United States is AASHTO's CoRe Structural Elements Guide. The guide is intended for national consistency in defining and inspecting CoRe elements. The major CoRe elements are structural elements that are subelements of the three NBI classes (deck, superstructure, and substructure). They provide more detailed condition information than the NBI condition ratings. The purpose of the work reported in this paper is to provide a methodology to predict the federally required NBI condition ratings from the CoRe element condition data, which could be utilized both at the state and national levels for resource savings.

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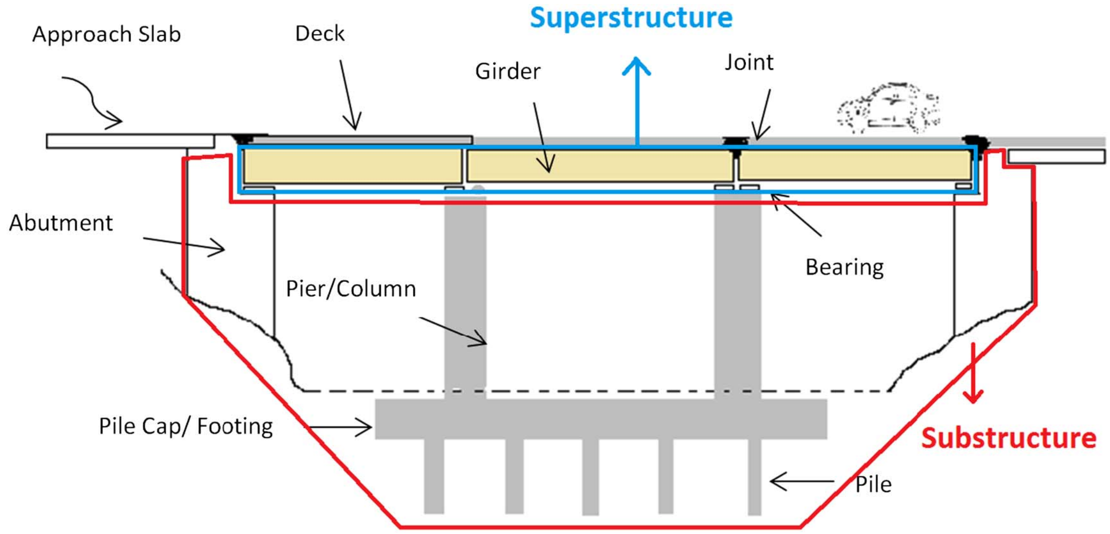

Fig. 1. Schema of major elements in a bridge

Background

For each CoRe element, there are three, four, or five condition states that represent different stages of deterioration. In CoRe element condition language, condition state one represents the best possible condition, while latter condition states represent worse. CoRe element-level inspection data are assigned in percentages, as relative quantities of a particular CoRe element in these condition states. Agencies can also define environments to indicate the severity of the external conditions for an element, or structural units for multiple-span structures. Therefore, the element condition data for one CoRe element can be fragmented across several structural units and environments. CoRe elements also include a set of special elements called smart flags, which allow tracking of distress conditions such as pack rust and deck cracking.

Table 1 gives an example Pontis inspection for a bridge in which there are two decks, one superstructure (in two environments), and six substructure elements. For element 275 (a reinforced concrete backwall) in Table 1, 72% of that element is in condition state 1 (which is the best condition), 25% of the element is in condition state 2, and 3% of the element is in condition state 3. The NBI condition ratings for this structure from the same field inspection are 6 (satisfactory) for the deck and 5 (fair) for both superstructure and substructure. While NBI condition ratings give an overall condition measure, they lack the precision and quantitative detail provided by the CoRe element condition data.

Review of Past Work

Because CoRe element condition data contain more detailed information on the general NBI classes, state agencies, experts in the field, and FHWA questioned the redundancy of collecting NBI condition ratings. As a response to this interest, Hearn et al. (1993) developed the *BMSNBI* (also known as the NBI Translator) software that maps CoRe element condition data to NBI condition ratings for FHWA.

The *BMSNBI* has been available as a built-in module within Pontis BMS since 1994. States can report translated NBI condition ratings instead of field ratings to FHWA for the annual NBI submissions. However, only three state agencies use *BMSNBI* for their NBI data submissions (Bektas 2011), mostly due to general skepticism among the states regarding the accuracy of the *BMSNBI* algorithm (Hale et al. 2007). In comparison studies between generated and field ratings (Herabat and Tangphaisankun 2005; Aldemir-Bektas and Smadi 2008), low predictive accuracies and different condition distributions at the network level were reported. *BMSNBI* is designed to be conservative in assigning NBI ratings and thus tends to assign lower ratings, especially for ratings above 7.

BMSNBI generates NBI condition ratings by a four-step process (Hearn et al. 1997). In the first step, CoRe elements of a bridge are grouped into matching NBI fields. In the second step, NBI ratings are assigned for each CoRe element based on *BMSNBI* maps. In the

Table 1. Example Pontis Bridge Inspection Data and Matching NBI Condition Ratings

Number	Environment	Element name	States	NBI subsystem	P_1	P_2	P_3	P_4	P_5	NBI rating
14	3	Concrete deck/Asphalt concrete overlay	5	Deck	0	0	100	0	0	6
110	2	RC open girder	4	Superstructure	88	12	0	0		5
346	3	RC girder end	4	Superstructure	72	11	6	11		
205	2	RC column	4	Substructure	38	31	31	0		6
215	3	RC abutment	4	Substructure	76	15	9	0		
234	1	RC cap	4	Substructure	63	20	17	0		
275	2	RC backwall	4	Substructure	72	25	3	0		
300	1	Strip seal expansion joint	3	NA	0	20	80	0		
301	1	Pourable joint seal	3	NA	100	0	0			
311	2	Moveable bearing	3	NA	0	100	0			
321	1	RC approach slab	4	NA	70	30	0	0		
331	1	Concrete bridge railing	4	NA	62.5	37.5	0	0		

Note: NA = not applicable.

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Requirements on Element Quantities			NBI Rating
P_1	$\geq$	$M_{1,9}$	9
$P_1 + P_2$	$\geq$	$M_{2,9}$	
$P_1 + P_2 + P_3$	$\geq$	$M_{3,9}$	
$P_1 + P_2 + P_3 + P_4$	$\geq$	$M_{4,9}$	
P_1	$\geq$	$M_{1,8}$	8
$P_1 + P_2$	$\geq$	$M_{2,8}$	
$P_1 + P_2 + P_3$	$\geq$	$M_{3,8}$	
$P_1 + P_2 + P_3 + P_4$	$\geq$	$M_{4,8}$	

Fig. 2. Part of a typical *BMSNBI* map

third step, NBI ratings for NBI elements are computed by weighted combinations of the NBI ratings assigned to the constituent CoRe elements. The weights are determined from the relative quantities and dimensions of elements by default. Equal weights and weights from gross quantities without separation by dimension are two other options (Hearn et al. 1997). In the last step, NBI ratings are modified by smart flags, if present.

BMSNBI maps are composed of a set of criteria for each NBI rating. Each criterion checks a sum of percentages in condition states of a CoRe element to a minimum required sum (Hearn et al. 1997). A typical form of the *BMSNBI* map is shown in Fig. 2.

Percentages of CoRe element quantities are denoted by P_i, where i denotes the ith condition state. Minimum required sums for each criterion, called mapping constants and denoted by $M_{i,j}$, were calibrated by exhaustive search procedures to arrive at an optimal set of mapping constants (Hearn et al. 1997) using data from 10 state DOTs. The four criteria for each NBI rating should simultaneously be satisfied for a particular CoRe element to be assigned to that particular NBI rating. After NBI ratings are assigned for all constituent CoRe elements, an NBI rating is computed for the specific NBI element and modified if necessary, following the final two steps of the *BMSNBI* process.

Besides its use for NBI submissions, the *BMSNBI* has two other major uses. The first use is within Pontis BMS modeling framework (Pontis 4.x) because the algorithm is used to develop performance measures for forecasted future conditions (*Pontis Bridge Management Release 4.4*). The second major use is within the National Bridge Investment Analysis (NBIAS) software tool, which is used by FHWA to forecast bridge needs for the condition and performance reports prepared for Congress. Therefore, the accuracy issues in the algorithm potentially affect not only the NBI submissions, but also Pontis BMS simulations and forecasted bridge needs reported to Congress.

Two recent studies, dated 2007 and 2008, proposed alternative translators. The results of this study are compared with these two studies in the "Discussion" section of this paper.

The first was an artificial neural network (ANN) model (Al-Wazeer et al. 2007), which achieved some improvement in prediction accuracy. In this ANN model, the single input vector contained all element-level data and the target output was the set of three NBI ratings.

The second study (Sobanjo et al. 2008) proposed another alternative tool called the NewTranslator. The modeling methodology for the NewTranslator is similar to that of Bridge Health Index (Shepard and Johnson 1999), which is an index calculation. NBI rating for an NBI class is calculated based on the individually assigned NBI ratings of CoRe elements and element weights based on expert opinion. Sobanjo et al. (2008) reported that the NewTranslator is more accurate in assigning NBI ratings in the higher range.

Table 2. Data Received from the Agencies

Agency	NBI condition ratings	CoRe element condition data	BMSNBI translated ratings
DOT_A	x	x	x
DOT_B	x	x	x
DOT_C	x	x	

Data

For the classification and regression tress (CART) analyses in this study, simultaneous CoRe element condition data and NBI condition ratings from three state DOTs archived for the 2006–2010 duration are used. None of these agencies use translated ratings for agency purposes or reports them to FHWA. However, for DOT_A and DOT_B, the input file for the *BMSNBI* was updated and translated ratings were obtained from both agencies for a set of observations for comparison purposes (Table 2). Some differences exist in bridge inspection and condition assessment practices across the agencies. The BMS implementation in DOT_A is at its earlier stages. DOT_A has a quality assurance (QA) process for the NBI condition ratings, but not yet for the CoRe element condition data.

State DOT_B and DOT_C are at similar stages of BMS implementation. Both states use CoRe element condition data for assessing their bridge conditions. They also make use of BMS decision support capabilities while they select bridge work candidates. They have QA reviews for CoRe element condition data. State DOT_B provides in-house trainings for bridge inspectors each year, while State DOT_C provides similar trainings every 2 years.

Methodology

The CART algorithm is a nonparametric data mining method, which is used for generating decision trees from a set of learning cases (Breiman et al. 1984; Vayssières et al. 2000). Classification tree algorithms have been applied to diverse problems such as remote sensing, character recognition, and medical diagnosis and prognosis (Gelfand et al. 1989). This technique is also commonly known as recursive partitioning because the algorithm recursively splits the predictor space by searching all possible splits along the range of predictor variables to grow a decision tree structure that best predicts the response variable. The response variable is usually either categorical (classification trees) or numeric (regression trees). The classification trees developed in this paper predict NBI condition ratings (response variable) based on the relative quantity of CoRe elements in their defined condition states (predictor variables).

Among a variety of classification tree algorithms that are available in the literature (Loh 2011), the CART algorithm implemented via *SAS JMP Pro* partition platform is used in this paper.

Classification tree algorithms are typically composed of three steps (De'ath and Fabricius 2000): growing trees based on splitting criteria, pruning trees, and selecting the optimal tree. Classification trees are grown by repeatedly splitting the data along only one explanatory variable at each split. Initially, the data are in one root node (node 1, Fig. 3). With recursive splits, the root nodes (leaves) are split into intermediary child nodes (nodes 2 and 3). Once the optimal tree is selected, the end nodes of the tree become the terminal nodes (nodes 4, 5, 6, and 7).

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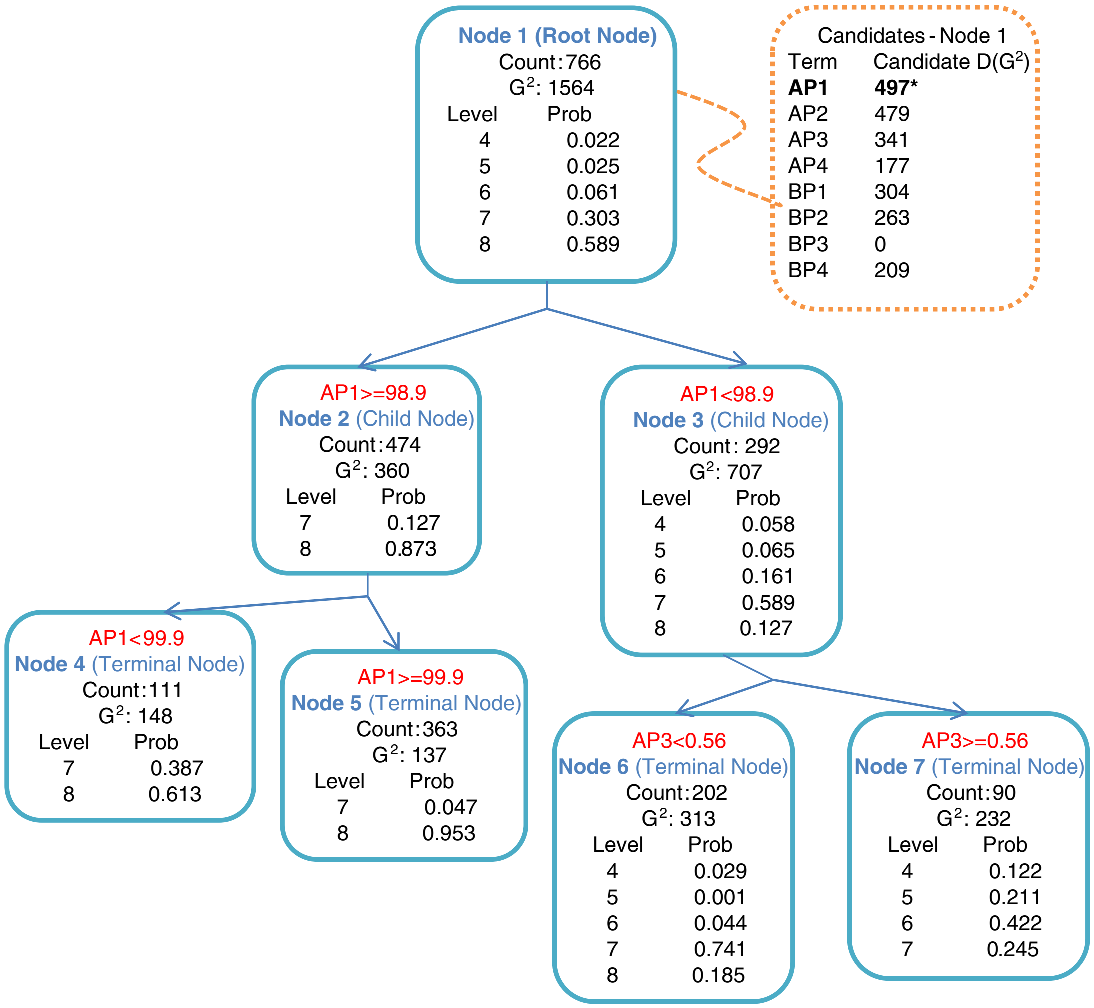

Fig. 3. Example classification tree

At each split, the data are partitioned into two groups (nodes), each of which is as homogeneous as possible with respect to the response variable. The homogeneity of nodes is measured by an impurity function, which is equal to zero for completely homogeneous nodes, with only one response level represented in the node. The value of the impurity function increases as the variability of the response in the node increases.

The CART algorithm uses a generalized version of the likelihood ratio statistic G^2 as the impurity function and as the splitting criterion. For a node, G^2 is twice the negative log-likelihood under the null hypothesis that says that the candidate node is homogeneous.

For node k

$$G_k^2 = -2\sum_{i=1}^{n} \ln(p_{ij/k}) \qquad (1)$$

where i = index of observations; n = total number of observations in node k; and $p_{ij/k}$ = proportion of observations of class j (e.g., inspected NBI condition rating) in node k. In Fig. 3, the proportion of observations for NBI condition classes 4–8 are given in node 1 (prob). Starting with the root node, the algorithm evaluates the goodness of candidate splits measured by the decrease in impurity, $D(G^2)$. The candidate $D(G^2)$ values in Fig. 3 show the highest possible decrease in G^2 for splits along the eight predictor variables. The point of best split along a specific predictor variable is also searched and identified by the algorithm. The decrease in impurity for a split s can be calculated as

$$D(G^2)_s = G^2_{\text{node } t} - G^2_{(\text{node } t)_{\text{right}}} - G^2_{(\text{node } t)_{\text{left}}} \qquad (2)$$

where $G^2_{\text{node } t} = G^2$ of the node to split; and $G^2_{(\text{node } t)_{\text{right}}}$ and $G^2_{(\text{node } t)_{\text{left}}} = G^2$ values of the resulting child nodes after the split.

The example tree in Fig. 3 is for predicting NBI substructure rating. $AP1 - AP4$ denotes the relative quantity of column elements (CoRe) in condition states 1–4, while $BP1 - BP4$ denotes the similar data for abutment elements. For node 1, predictor variable AP1 presents the highest candidate decrease in G^2. The sum of the G^2 values for nodes 2 and 3 after the split is 497 less than the G^2 value of node 1. Further splits that grow a classification tree gradually reduce the overall G^2. After each split, the algorithm fits the predicted probabilities for each response level for each node (Fig. 3).

CART is prone to overfitting by growing on noise; therefore, trees should be refined to better report the predictive ability of the trees. Breiman et al. (1984) and De'ath and Fabricius (2000) introduce three alternatives for selecting the best tree. The first idea is tree pruning by growing an overlarge tree and then cutting it back. This method can be infeasible because the number of splits can be very large. The second alternative is to find a sequence of nested trees of decreasing size, each of which is the best of all trees in the same size, using the overall misclassification rate. The final alternative is selecting tree size by cross validation and using honest estimates of error to select the optimal tree. Cross validation was utilized in this paper.

Each data set was divided into training and validation sets; one-third of each data set was randomly assigned as the latter. The stopping criterion was based on the entropy R-square. If the validation entropy R-square could not be improved with the next 10 splits, the splitting was stopped. At this point, the class with the highest predicted probability in each end leaf becomes the most likely class and assigned as the predicted NBI condition class. Thus, for each observation, the algorithm predicts a probability for each response variable class, and the sum of these predicted probabilities is equal to 1. The prediction formula is composed of nested conditional clauses that describe a tree structure.

Two coefficients of determination are reported as measures of the predictive ability of the classification trees in the partitioning platform. The first one, entropy R-square, compares the log-likelihoods from the fitted model (final tree) and the constant probability (null) model (distribution of the response variable in the root node), as in Eq. (3)

$$\text{Entropy } R-\text{square} = 1 - \left\{ \left[\sum_{i=1}^{n} \ln(p_{m,i}) \right] \left[\sum_{i=1}^{n} \ln(p_{0,i}) \right]^{-1} \right\} \tag{3}$$

where i = index of observations; n = total number of observations in the data set; $p_{m,i}$ = predicted probability of the assigned response level for each observation; and $p_{0,i}$ = proportion of the assigned response level in the root node. The second measure, generalized R-square, is a generalization consistent with the classical R-square, which is asymptotically independent of the sample size (Nagelkerke 1991). Generalized R-square can be interpreted as the proportion of the explained variation by the model and calculated as in Eq. (4)

$$\text{Generalized } R-\text{square} = 1 - \left[\frac{\sum_{i=1}^{n} \ln(p_{0,i})}{\sum_{i=1}^{n} \ln(p_{m,i})} \right]^{2/n} \times \left\{ 1 - \left[\sum_{i=1}^{n} \ln(p_{0,i}) \right]^{2/n} \right\}^{-1} \tag{4}$$

Depending on the type and length of bridge structures, the numbers of CoRe elements that constitute an NBI class vary. NBI deck elements, however, have only one matching CoRe deck or slab element. Other deck-related CoRe elements, bridge railings, and deck joints, are not to be considered in the overall NBI deck evaluation (FHWA 1995) and were thus not included in the analyses. Because the CoRe deck elements have five condition states, the predictor variables are the relative quantities of deck elements in these five condition states. These predictor variables are notated as $P1$ to $P5$ where index number represents the respective condition state. Unlike other CoRe elements, a deck element can be in only one of the five condition states, as a whole (a 100% in one of the condition states from 1 to 5). Consequently, for only decks, the algorithm predicts a nine-level response variable from only five cases of predictor variables.

The predictor variables for superstructures and substructures were assigned according to the structural function of the CoRe elements, regardless of the material. The superstructure for a smaller bridge is typically composed of a girder or a beam. Either CoRe element, independent of material property, bears the same type of load in the bridge. As the span length increases, a variety of other BMS superstructure elements are seen in the design such as additional beams, girders, girder ends, frames, hinges, or bearings. The notation of predictor variables for one-element superstructures or substructures is similar to that of deck elements, but for only four condition states, $P1$ to $P4$. When there is more than one CoRe element in a superstructure or substructure, the notation of predictor variables were preceded with letters A, B, C, or D. For instance, for a two-element tree the predictor variables were notated as $AP1 - AP4$ for the first element and $BP1 - BP4$ for the second element.

The classification trees modeled in this paper typically had up to three CoRe elements, with the exception of the four-element substructure tree for DOT_A (Table 3). While structures with higher number of CoRe elements were observed, the 20 predictive trees represented the general cases and 94–97% of the observations of the three data sets.

Three typical combinations that represented NBI substructures were as follows: a single abutment element (13% of all inspections), a combination of one span support element (e.g., a wall, column, pier, or pile) with an abutment element (26% of all inspections), a cap element to complement one span support element, and an abutment (53% of all inspections).

Results

This section presents the accuracy and reliability of the predictions by reporting R-square values, percentage of exact predictions and misclassification rates, ranges of predicted probabilities, and column contributions.

The generalized R-square values [Eq. (4)] for the final predictive trees vary from 45 to 84% and are typically more than 60% for superstructure and substructure ratings (Table 3). The values indicate that the predictive trees explain the variation in the data substantially well. Because the predictor variables for the deck elements are limited in variability, the lower R-square values for deck trees are expectedly lower. Entropy R-square values for training and validation data sets (R-square T and R-square V) are quite close with the exception of three data sets with fewer numbers of observations (trees 11, 13, and 15 in Table 3).

Because the CART technique is a nonparametric method, the predictions are not accompanied by reliability measures such as confidence intervals or standard errors of prediction. The predicted probability for each NBI condition class, among nine candidate classes, determines the predicted NBI condition class for each observation. Consequentially, the ranges of these probabilities are evaluated as an intuitive measure of reliability. If a predicted probability is higher than 0.5, no other candidate class among the other eight candidate classes may have a higher predicted probability, and that specific class is dominantly the predicted class. Table 4 presents the frequency of the predicted probabilities for assigned classes with five ranges. The predicted probabilities below 0.4 shows that the real NBI condition classes in the end leaves of the predicted trees are more heterogeneous. Among the predictions for decks, the DOT_A tree, with a majority of the predicted probabilities below 0.4, presents weaker predictions. Despite the limited nature of the predictor variables for decks, the predicted NBI deck classes for DOT_B and DOT_C still have high probabilities. Among all agencies, for substructure and superstructure ratings DOT_A has more frequent observations in the lower probability ranges. The majority of the predictions for DOT_B and DOT_C are high-probability estimates, with exceptionally high probabilities for DOT_B, which are higher than 0.6 for more than 85% of the observations.

The predictive accuracy of the trees is evaluated by assessing the misclassification rates. In Table 5, the ratio of correct predictions and misclassified observations by agency and NBI class type are

Table 3. Classification Trees Summary

Number	NBI	DOT	Number of elements	Elements	Condition state	Number of splits	Number of observations	R-square T	R-square V	General R-square
1	Deck	A	1	Deck/slab	5	4	3,218	0.23	0.2	0.54
2	Deck	B	1	Deck/slab	5	4	7,570	0.31	0.31	0.54
3	Deck	C	1	Deck/slab	5	4	7,045	0.23	0.21	0.45
4	Superstructure	A	1	Girder/beam	4	6	2,532	0.26	0.24	0.59
5	Superstructure	B	1	Girder/beam	4	16	2,241	0.64	0.56	0.84
6	Superstructure	C	1	Girder/beam	4	13	7,299	0.32	0.3	0.6
7	Superstructure	B	2	Girder/beam and bearing/girder end/hinge	4 + 3	12	1,143	0.6	0.54	0.81
8	Superstructure	C	2	Girder/beam and second girder/beam OR frame	4	13	840	0.34	0.26	0.63
9	Superstructure	B	3	Girder/beam, bearing, and second bearing/hinge	4 + 3 + 3	11	1,101	0.4	0.29	0.68
10	Substructure	A	1	Abutment	4	3	300	0.26	0.21	0.59
11	Substructure	B	1	Abutment	4	27	336	0.51	0.19	0.8
12	Substructure	C	1	Abutment	4	14	2,146	0.37	0.29	0.65
13	Substructure	A	2	Pile/column and abutment	4	11	162	0.42	0.1	0.76
14	Substructure	B	2	Column and abutment	4	47	4,169	0.42	0.35	0.64
15	Substructure	C	2	Cap and abutment (part 1)	4	15	220	0.52	0.26	0.82
16	Substructure	C	2	Pile/column and abutment (part 2)	4	5	668	0.26	0.24	0.51
17	Substructure	A	3	Pile/wall/column, abutment, and cap	4	17	1,991	0.25	0.15	0.52
18	Substructure	B	3	Column, abutment, and cap	4	27	3,123	0.47	0.44	0.71
19	Substructure	C	3	Pile/wall/column, abutment, and cap	4	43	5,634	0.36	0.31	0.65
20	Substructure	A	4	Cap, abutment, backwall, and wall/Column	4	17	1,514	0.29	0.19	0.63

Table 4. Percentage of CART Predictions by Predicted Probability Ranges

NBI subsystem	Predicted probability	DOT_A (%)	DOT_B (%)	DOT_C (%)
Deck	<0.2	0	0	0
	[0.2–0.4)	88	0	0
	[0.4–0.6)	12	76	1
	[0.6–0.8)	0	8	99
	≥0.8	0	15	0
Substructure	<0.2	0	0	0
	[0.2–0.4)	8	0	2
	[0.4–0.6)	78	9	27
	[0.6–0.8)	13	49	57
	≥0.8	1	41	15
Superstructure	<0.2	0	0	0
	[0.2–0.4)	13	0	0
	[0.4–0.6)	87	14	64
	[0.6–0.8)	0	33	35
	≥0.8	0	53	1

presented. Misclassification error is defined as the difference between the field rating and the predicted rating (inspected NBI condition rating—CART predicted NBI condition rating). Therefore, positive values indicate cases in which predicted ratings were lower than real ratings and negative values indicate higher predictions. As expected, correct matches for deck ratings are lower than the other two rating types but correspond to 39, 63, and 70% of all observations for DOT_A, DOT_B, and DOT_C, respectively. Across agencies, DOT_A has lower accuracies; the correct matches for the other agencies range from 60 to 80%. With the exception of decks, more than 94% of all observations are predicted within one misclassification error.

Figs. 4–6 show a visual comparison of the correct predictions and misclassifications by agency and rating type. The frequencies of class predictions that are one class higher or lower are typically close across agencies, but for DOT_B decks one class higher predictions are 29% more than one class lower predictions. A similar exception is observed for DOT_C superstructure ratings, where

Table 5. CART Accuracy

Error[a]	Deck DOT_A	Deck DOT_B	Deck DOT_C	Substructure DOT_A	Substructure DOT_B	Substructure DOT_C	Superstructure DOT_A	Superstructure DOT_B	Superstructure DOT_C
−3 (%)				0.2					
−2 (%)	7.9		0.5	0.9	0.3	0.8	1.2	0.3	0.8
−1 (%)	21.3	33.1	15.3	19.8	6.7	19.1	25.2	6.4	27.1
0 (%)	39.0	62.7	69.6	54.0	78.4	64.6	48.7	81.0	59.6
1 (%)	22.2	3.9	12.5	20.7	13.2	13.5	19.8	10.1	11.6
2 (%)	8.6	0.3	1.7	3.5	1.1	2.0	4.2	1.6	0.7
3 (%)	1.0	0.1	0.4	0.8	0.2	0.1	0.8	0.6	0.1
4 (%)				0.1	0.1				
Error = 0 (%)	39.0	62.7	69.6	54.0	78.4	64.6	48.7	81.0	59.6
|Error| <= 1 (%)	82.5	82.5	97.4	94.5	98.3	97.2	93.8	97.5	98.3
|Error| <= 2 (%)	99.0	99.0	99.6	98.9	99.7	99.9	99.2	99.4	99.9
|Error| <= 3 (%)	100	100	100	99.87	99.95	100	100	100	100
Number of observations	3,218	7,570	7,045	3,967	7,628	8,668	2,532	4,485	8,139

[a]Error = Inspected NBI condition rating—CART predicted NBI rating.

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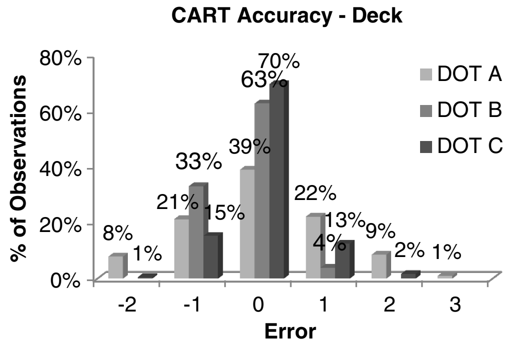

Fig. 4. Distribution of misclassification error for CART deck ratings

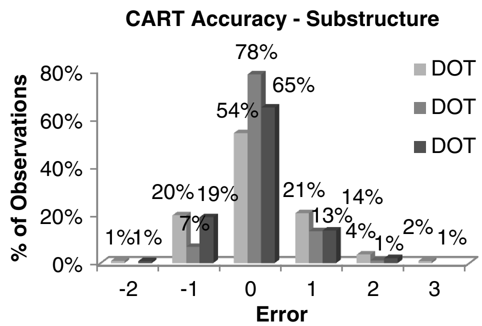

Fig. 5. Distribution of misclassification error for CART substructure ratings

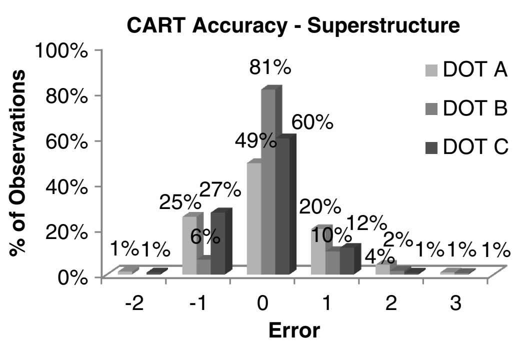

Fig. 6. Distribution of misclassification error for CART superstructure ratings

one class higher predictions are 15% more than one class lower predictions.

The influence of the predictor variables on the NBI class prediction is examined with the help of the column contributions, which are presented in Figs. 7–12. Measured as a percentage of the initial G^2 reduced by the splits on that particular predictor variable, relative magnitudes of column contributions indicate relative importance of the variables in the trees. Column contributions provide a measure of relative importance for both the condition states within a CoRe element and the different CoRe elements that constitute an NBI class. Without exception, the relative quantity of CoRe elements in the first condition state (i.e., $AP1$, $BP1$, or $CP1$) is the most important predictor variable among other condition states. For trees modeled for bridges that have one CoRe element per NBI class (Figs. 7–9), more than 60% of the models were influenced by condition state 1. The underlying reason is interpreted to be the typical distribution of relative quantities across

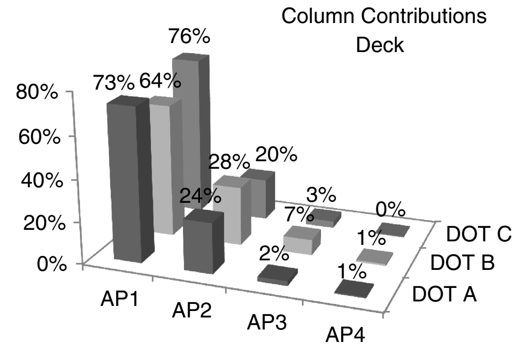

Fig. 7. Column contributions for CART predictions for deck ratings

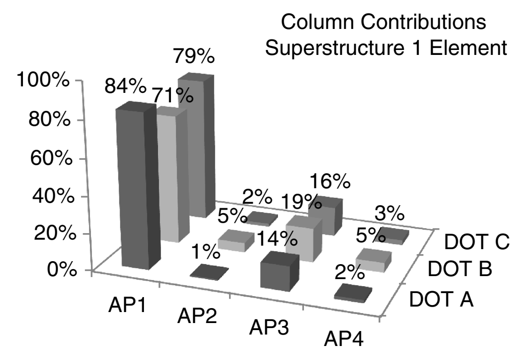

Fig. 8. Column contributions for CART predictions for superstructure ratings, one CoRe element

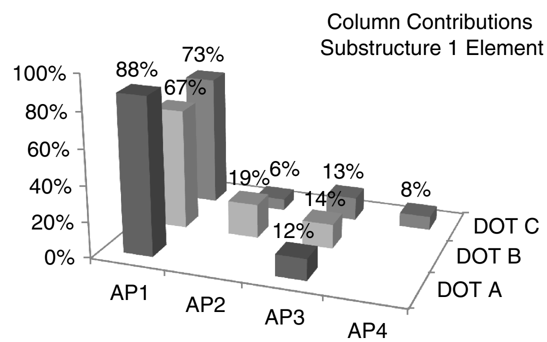

Fig. 9. Column contributions for CART predictions for substructure ratings, one CoRe element

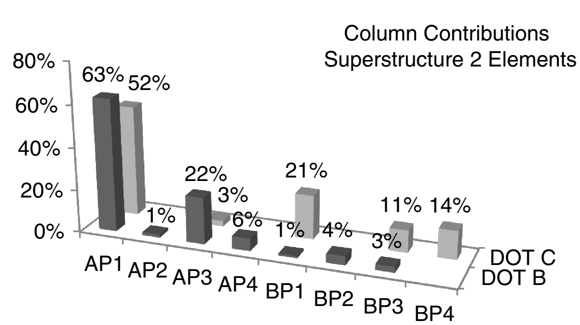

Fig. 10. Column contributions for CART predictions for superstructure ratings, two CoRe elements

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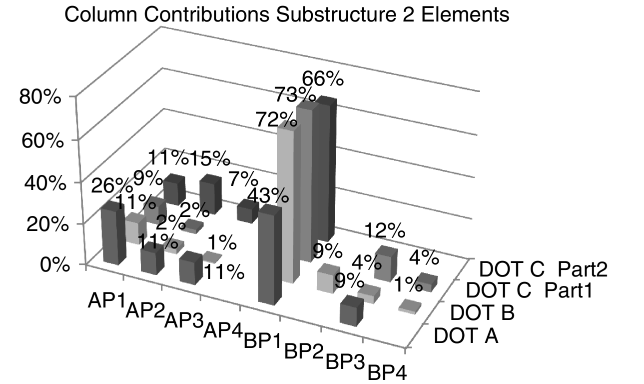

Fig. 11. Column contributions for CART predictions for substructure ratings, two CoRe elements

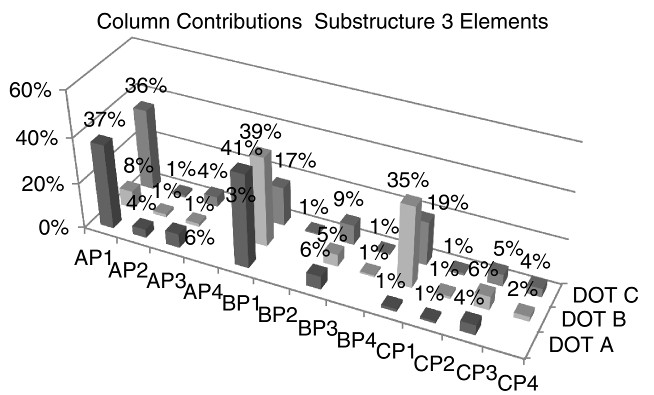

Fig. 12. Column contributions for CART predictions for substructure ratings, three CoRe elements

condition states. Among the condition states, the first condition state is where the most variability in the relative quantity of a CoRe element is observed and the distribution is skewed toward higher values.

The distribution of column contributions for models with one CoRe element is quite similar across agencies, except for the differences in the second and fourth condition states for substructures. When the superstructures are examined with two CoRe elements, the second elements for DOT_C contribute to the model almost as much as the first elements (Fig. 10), while the contributions of the second elements for DOT_B are minimal. This observation is logical from a structural perspective because the second element for DOT_C is a second girder or a beam element (Table 3), which is a primary structural element, whereas the second element for DOT_B is a bearing or girder end. Abutments dominantly influence the two-element models for substructure models as the second elements (Fig. 11). Cap elements assigned as the first elements for the DOT_C part 1 model have a small contribution of 11% when compared with higher contributions of pile or column elements.

Among three-element substructure models, the column contributions for element types vary. The column elements (AP_{1-4}) have fewer and the cap elements have more contributions in tree development in comparison with the other agencies. Unlike others, the data set for DOT_B has only column elements as the first element versus pile, column, or wall elements for DOT_A and DOT_C. The models for DOT_B and DOT_C have higher generalized R-square values and more contribution from the cap elements when compared with the DOT_A model.

Discussion

A direct comparison of *BMSNBI* translated ratings versus CART predicted ratings for DOT_A and DOT_B indicate substantial improvement in predictive accuracy for the CART methodology (Table 6). While the increases in exact predictions for DOT_A are only 6, 24, and 2% for deck, substructure, and superstructure ratings, respectively, corresponding increases were 14, 39, and 56% for DOT_B. More than 98% of all predictions for DOT_B are within one error term, except decks.

When predictive accuracy is compared across agencies, it can be observed that the predictions from DOT_A are not as accurate as the predictions from the other agencies. Unlike DOT_B and DOT_C, DOT_A does not have QA reviews for the CoRe element condition data and does not utilize CoRe element condition data for assessing bridge conditions. It can be inferred that these differences affect the quality of the CoRe data, the consistency between the two rating methodologies, and eventually the accuracy of prediction.

Analysis on the column contributions indicates higher influence of first condition state and primary structural elements such as abutments for substructures and girders or beams for superstructures on

Table 6. CART versus *BMSNBI* Comparison

	Deck				Substructure				Superstructure			
	DOT_A		DOT_B		DOT_A		DOT_B		DOT_A		DOT_B	
Error[a]	CART	*BMSNBI*	CART	*BMSNBI*	CART	*BMSNBI*	CART	*BMSNBI*	CART	*BMSNBI*	CART	*BMSNBI*
−4 (%)		0.2				0.1		0.1		0.1		
−3 (%)		1.9		0.1	0.3	1.0		0.2		1.5		0.1
−2 (%)	10.6	20.0		1.0	0.9	13.6	0.2	1.9	1.1	5.0	0.3	21.2
−1 (%)	23.1	45.8	33.2	49.4	21.5	42.8	7.4	54.6	28.8	31.4	6.3	47.3
0 (%)	**38.2**	**32.1**	**62.7**	**48.5**	**53.4**	**28.9**	**79.5**	**40.7**	**46.5**	**45.1**	**82.6**	**27.2**
1 (%)	23.2		3.8	0.8	20.3	10.9	12.1	2.4	19.4	14.2	9.0	3.8
2 (%)	4.7		0.2	0.1	2.8	2.4	0.7	0.1	3.5	2.5	1.3	0.4
3 (%)	0.2				0.7	0.3	0.1		0.8	0.2	0.5	0.1
4 (%)					0.2							
Number of observations	1,287		2,881		1,938		2,877		1,179		1,693	
Error = 0 (%)	38.2	32.1	62.7	48.5	53.4	28.9	79.5	40.7	46.5	45.1	82.6	27.2
|Error| <= 1 (%)	84.5	77.9	99.7	98.8	95.1	82.7	99.0	97.7	94.7	90.8	97.9	78.3
|Error| <= 2 (%)	99.8	97.8	100.0	99.9	98.8	98.7	99.9	99.7	99.2	98.2	99.5	99.9
|Error| <= 3 (%)	100.0	99.8	100.0	100.0	99.8	99.9	100.0	99.9	100.0	99.9	100.0	100.0

[a]Error = Inspected NBI condition rating—CART predicted NBI rating.

developing trees. Despite some variability across agencies, column contributions, together with tree structure, can be utilized in future work to refine trees from different agencies in order to develop national level models.

Conclusions

This paper develops a new methodology to predict NBI condition ratings from CoRe element condition data. Statistical results propose a potentially more accurate method than the previous algorithms in the literature. Direct comparison of predictions from the CART algorithm and the *BMSNBI* indicates better accuracy of the CART algorithm. The CART algorithm also achieved higher accuracies than two other alternative methods in the literature when similar accuracy measures were compared.

The proposed methodology does not make assumptions about the impacts (weights) of specific CoRe elements on the related NBI condition ratings. On the contrary, due to the way the predictor variables were defined, the column contributions from the CART results suggest the statistical impact of a specific CoRe element type and condition state in the prediction. Analyses of such information can be useful to state transportation agencies for exploring how the condition of different CoRe elements contributes to the assignment of NBI condition ratings.

The statistical results from this paper and the classification trees from the CART algorithm are specific to the states in this paper and cannot be generalized. Yet, the paper provides a prediction methodology based on simple logical conditions that can be used to create easy-to-understand business rules for transportation agencies. Future work supported with the collaboration of other DOTs can develop refined national level trees that can be utilized in condition assessment and decision support tools at the state and national level.

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